

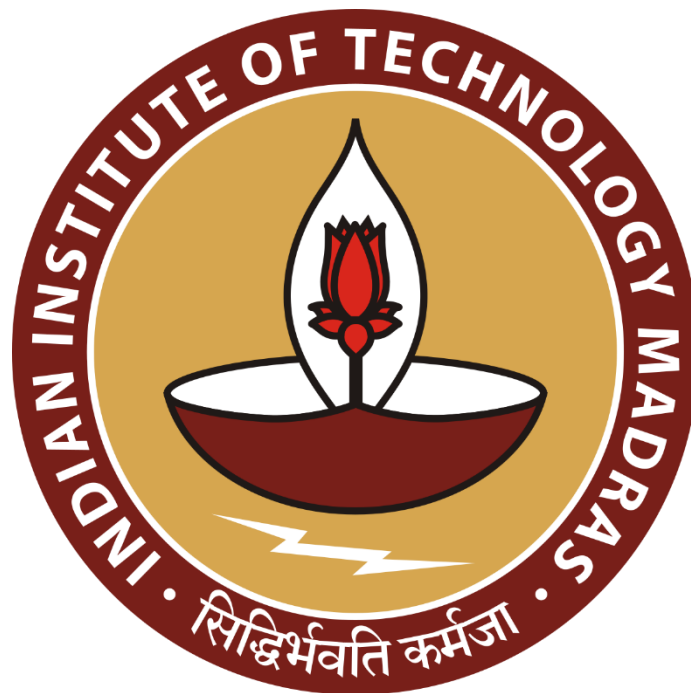
Enhancing Gym Profitability by Optimizing Member Retention & Acquisition

Final Report for the BDM capstone Project

Submitted by:

Name: DAKSH

Roll number: 23F2001398



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Declaration Statement

I am working on a Project titled **“Enhancing Gym Profitability by Optimizing Member Retention & Acquisition”**. I extend my appreciation to **F4 Fitness Gym**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: Daksh

Name: Daksh

Date: 10/12/2024

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Executive Summary and Title

This report is the final report for the BDM Capstone Project, the objective of this project is to enhance the profitability of F4 Fitness, a gym located in Pochanpur, Sector –23, Dwarka, Delhi, which addresses two main business challenges: member retention and acquisition, and managing overcrowding during peak hours. The purpose of the analysis is to provide insights and recommendations to address these challenges and promote business growth.

The data was collected for eight months (January-August 2024) and analyzed using organization's demographics data, member behavior data, member distribution, overcrowding data. The analysis was standardized using various methods, including line charts for membership trends, pie charts for gender distribution, and column charts for time slot usage. In the report time slot analysis, trends related to gender ratio, membership type, and equipment's usage were identified.

The main conclusions indicate that the number of members had increased in the months of March, June, and July, while the number of members was affected during months of April and May due to the re-opening of schools and colleges.

The recommendations include giving discounts to members during off-peak hours, increasing the number of trainers during peak hours, starting a loyalty program for short-term members, and extending the gym's operating hours.

By implementing data-driven recommendations, we can improve crowd and member acquisition and retention, which will boost member satisfaction and increase profitability for F4 Fitness. This ensures sustainable growth and a better member experience.

Detailed Explanation of Analysis Process/Method

1. Data Collection

The data for this analysis was collected from F4 Fitness Gym's record book over a period of eight months (January-August 2024). The data collection process focused on key areas to ensure a comprehensive understanding of member demographics, behavior, distribution patterns, and crowd interactions.

Demographics Data

- **Timeframe:** Data collected for a 1-month period (i.e. August).
- **Description:** This dataset includes detailed information about the gym's members collected from admission forms.
- **Fields:**
 - ~ **Member ID:** A unique identifier to track activities for each member and it is allotted by the gym owner randomly.
 - ~ **Membership Type:** Information about the types of membership plans (e.g. monthly, quarterly, annual) selected by the member.
 - ~ **Age:** Member's age for assessing the age group of members.
 - ~ **Gender:** The gender of the member for assessing gender distribution.
 - ~ **Address:** General location data to find out the distance of gym from their residences.
 - ~ **Distance from Gym (in km):** The distance between the gym and the members' residences can provide insight into how far the members are willing to travel.
 - ~ **Fees (in Rupees per month):** Membership fees paid by each member, which can reveal pricing trends and the popularity of certain membership plans.

Member Behavior Data

- **Timeframe:** Data collected for 1 month period (i.e. August) and the check-in and check-out times are not date wise, they are the approximate times of their entries which were asked at the time of issuing membership.
- **Description:** This dataset focuses on members' gym usage patterns, such as when they go and how long they stay.
- **Fields:**
 - ~ **Member ID:** It is allotted by the gym owner randomly. Linked with demographics data to maintain consistency across datasets. It is collected for the month of August and it changes every month and number of males and females fluctuate every month.
 - ~ **Check-In Time:** The time members entered the gym.

- ~ **Check-Out Time:** The time members left the gym.
- ~ **Duration (hrs.):** The approx. time spent by members during each visit

Member Distribution Data

- **Timeframe:** Data collected for 8 months (January–August 2024).
- **Description:** This dataset highlights the main trends in membership distribution, focusing on gender and membership types during the 8 months period.
- **Fields:**
 - ~ **Member’s Gender Distribution Over Months:** Monthly records of the number of males, female members and number of males and females fluctuate every month.
 - ~ **Membership Subscription Type Distribution:** Data on different subscription plans, such as monthly, quarterly and annual.

Overcrowding Data:

- **Timeframe:** Data collected for 1 month period (i.e. August).
- **Description:** This dataset focuses on identifying crowd-related issues and optimizing the use of space and equipment.
- **Fields:**
 - ~ **Time Slot:** Time intervals (e.g., 6–7 AM, 7–8 AM) when data was collected.
 - ~ **Area/Zone:** Specific zones within the gym (e.g., cardio section, weightlifting area, etc.).
 - ~ **Equipment in Use:** The number and type of equipment in use during each time slot.
 - ~ **Number of Members:** Approx. number of members present during each time slot in every area.
 - ~ **Comments:** Members feedback on factors such as availability of tools, waiting time.

2. Data Cleaning

The data has been cleaned to ensure the exactness and stability of the analysis because the record was not maintained properly.

- **Duplicate Entries:** The additional member record has been removed to avoid distortion in the results.
- **Unnecessary Data:** Irrelevant columns have been filtered.
- **Tools used:** Microsoft Excel was used to collect and clean the data.

3. Data Analysis and Management

- **Membership Subscription Trends:** Managed data based on membership subscription type (e.g. monthly, quarterly, annual) to analyze retention.
- **Peak Hour Analysis:** Analyzed the data with check – in time to find out the peak hours of the day.
- **Peak Months of Business:** Analyzed the data of Member’s Distribution over months to find out the peak months of the year.

4. Analysis Tools and Techniques

- **Line Chart:**
Line chart representing the Member’s Distribution over a period of 8 months to find out the peak months of the business.
- **Pie Chart:**
Pie Chart representing Gender Ratio of Members for the Month (August)
- **Column Chart:**
Chart representing the Distribution of Members According to Check-in Time to find out the peak hours and non-peak hours of the day to solve the overcrowding problem at certain period of times.
- **Pie Chart:**
Pie- Chart showing distribution of members during peak and off - peak hours.

5. Analysis Steps

Line chart - Membership Subscription Type Distribution

- **Objective:** To identify the preferred membership subscription type over time.
- **Steps:**
 1. Collect data for each subscription type (e.g., monthly, quarterly, annual) over a defined period of time.
 2. Create a line chart with the x-axis representing the time period and the y-axis representing the number of subscriptions for each type.
 3. Use separate lines to represent each subscription type to easily compare trends.
- **Outcome:** This analysis will help determine if there are any clear preferences or trends in subscription types over time, which could potentially highlight periods of higher adoption for specific plans.

Pie Chart - Gender Ratio of Members for the Month (August)

- **Objective:** To clarify the distribution of genders among members for a specific month (August).
- **Steps:**
 1. Include only the data from the month of August in the given dataset.
 2. Count the total number of male and female members.
 3. Represent these calculations as a percentage of total membership for the month.
 4. Create a pie chart representing each gender (such as male, female), along with relevant labels and percentages.
- **Outcome:** The pie chart will visually present the gender ratio, which will provide insight into the diversity of August members based on demographics.

Line Chart - Member's Gender Distribution over a period of 8 months

- **Objective:** To examine trends and patterns in gender distribution within an eight-month period.
- **Steps:**
 1. Group the data based on month and gender.
 2. Count the total number of members for each gender on a monthly basis.
 3. Plot the line chart with the x-axis representing months and the y-axis representing the total number of members.
 4. Use separate lines to represent each gender, making comparisons clear.
- **Outcome:** This visualization pattern will reveal trends such as gender-specific development or fluctuations.

Column Chart - Distribution of Members According to Check-in Time

- **Objective:** To identify the peak hours for member check-ins during the day.
- **Steps:**
 1. Extract data related to member check-in time.
 2. Group the data by time intervals (e.g., hourly)
 3. Count the number of members for each time interval
 4. Create a column chart with the x-axis representing the time intervals and the y-axis showing the number of check-ins.
- **Outcome:** This chart will clearly display the peak hours of the day, which will assist in resource allocation and planning activities during peak times.

Results and Findings

1. Analysis of Membership Trends and Retention

1.1 Membership Subscription Type Distribution

The analysis of membership data involved a month – wise examination of F4 Fitness membership data from January 2024 to August 2024 and creating graphical representation to visually observe the subscription rates.

Month	Male	Female	Total	Membership Type		
				Monthly	Quarterly	Annual
January	81	35	116	98	14	4
February	75	42	117	98	15	4
March	83	40	123	102	16	5
April	74	30	104	89	10	5
May	73	35	108	92	11	5
June	90	44	134	118	11	5
July	85	45	130	116	9	5
August	82	38	120	106	9	5

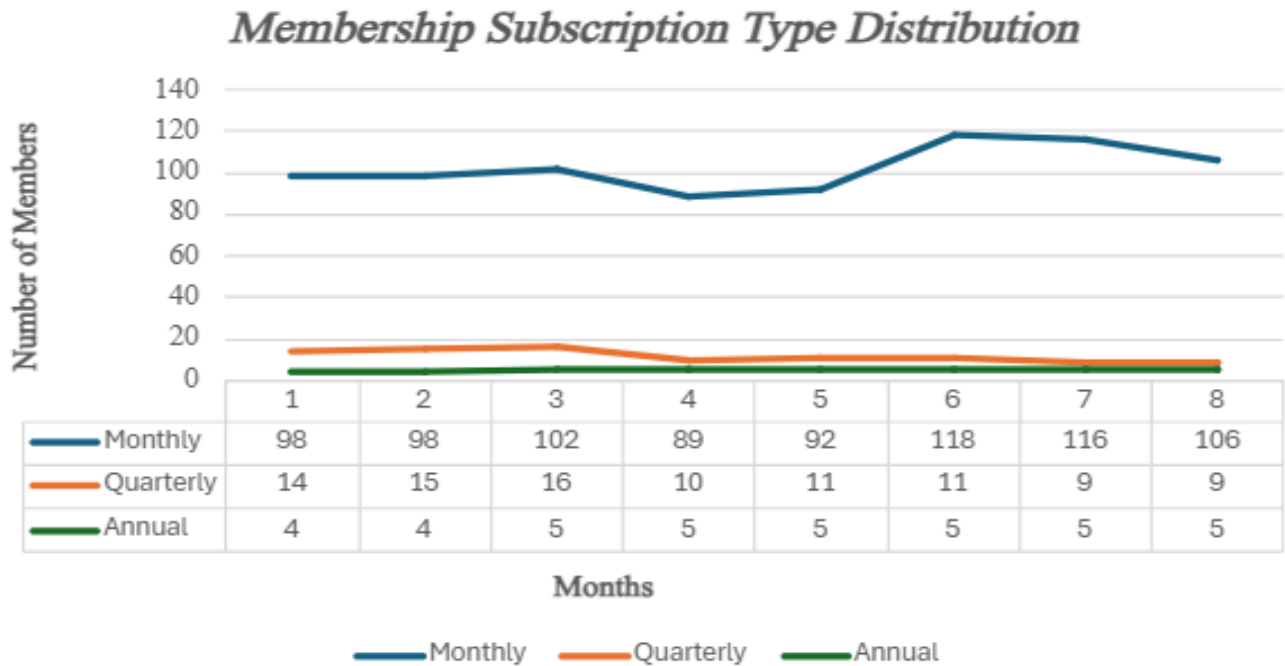


Fig 1. Line Chart for Membership Subscription Type Distribution over period of 8 months (January 2024 – August 2024)

- **About the Graph:** The above chart represents the Membership Subscription Type Distribution over period of 8 months (January 2024 – August 2024). Fee collection over period of 8 months has been calculated from this graph, and it has been determined which type of membership has benefited the gym the most (monthly, quarterly, or annually), and which type of membership is preferred by the members.
- **Findings:**
 1. **Fee Collection:** The total revenue generated from memberships during the period was **INR 11,61,600**.
 2. **Popular Membership Type:** Monthly membership is the most preferred option among members. However, this type of membership also has the highest dropout rate.
 3. **Revenue Insights:** Despite its high dropout rate, monthly membership contributes significantly to the gym's revenue due to its popularity. It indicates a continuous adoption by short-term members to maintain income levels.

1.2 Member Retention Patterns

Retention patterns were analyzed across different membership types to find out which subscription plans boost member loyalty.

- **Findings:**

1. **Higher Retention Rates:** Members with annual or quarterly membership subscriptions demonstrate higher retention rates compared to those with monthly subscriptions.
2. **Dropout Patterns:** Monthly membership, while popular, is more suitable for short-term commitment. Members who choose this plan often cancel after a few months, possibly due to short-term fitness goals or lack of long-term commitment.

2. Analysis of Overcrowding Data

Member Distribution Across Time Slots

Distribution of Members

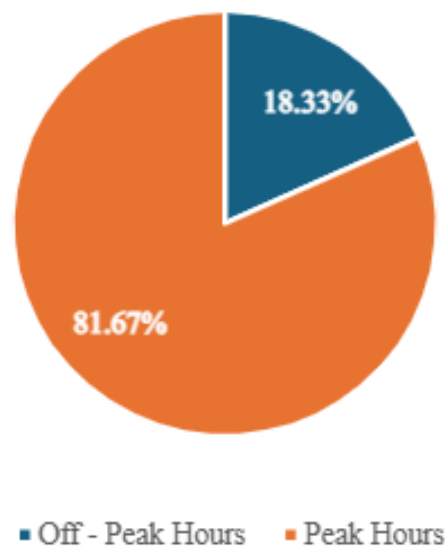


Fig 2. Pie- Chart showing distribution of members during peak and off - peak hours

- **About the Chart:** This chart illustrates the ratio of participating members in the gym during peak and off-peak hours. It provides a clear perspective on the distribution of members, highlighting their concentration during specific time periods.
- **Findings:**
 1. **Peak Hours Domination:** 81.67% of gym visits occur during peak hours.
 2. **Low Utilization during Off – Peak hours:** Only 18.33% of gym visits occur during off-peak hours.

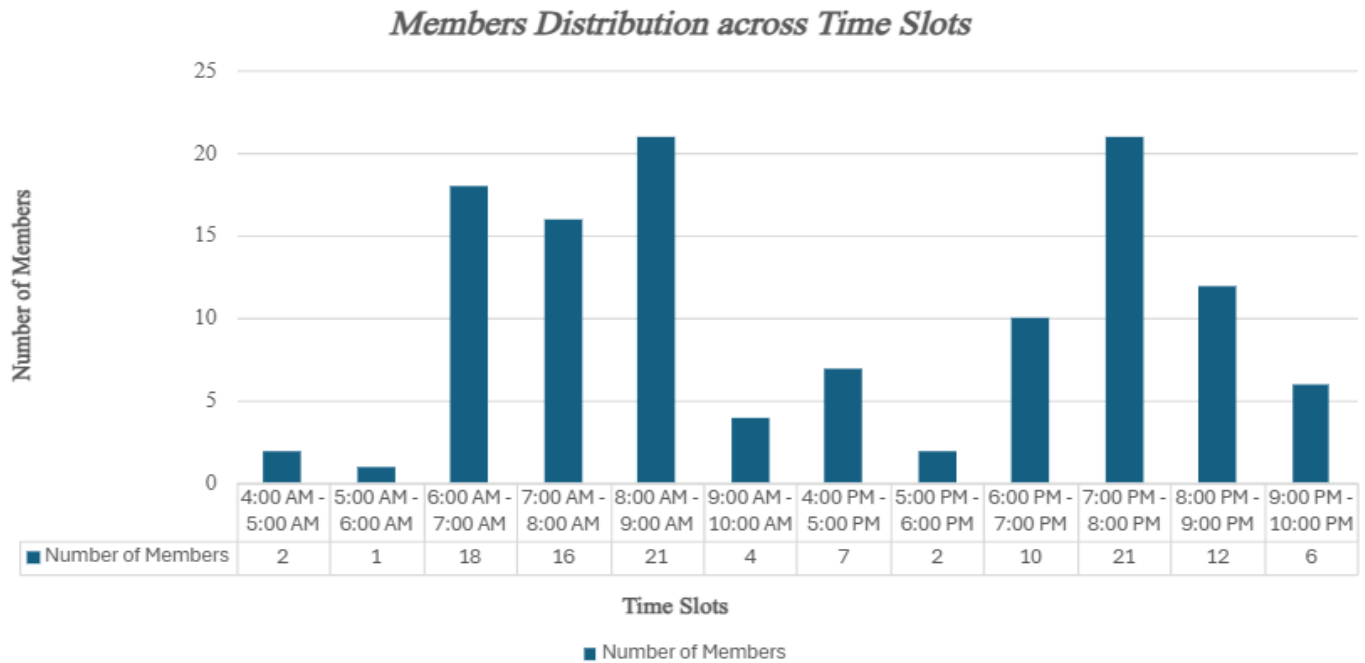


Fig 3. Column chart showing member distribution across time slots.

- About the Graph:** The above column chart represents the member distribution across time slots during day and this graph indicate the times during morning and evening operating hours when the gym experiences overcrowding.
- Findings:** Gym experiences overcrowding during certain periods of time in morning (6 AM – 9 AM) and certain period of times in evening (6 PM – 9 PM) and off-peak hours are (4 AM – 6 AM), (9 AM – 10 AM) in the morning and (4 PM – 6 PM), (9PM – 10 PM) in the evening.

3. Analysis of Gender Distribution

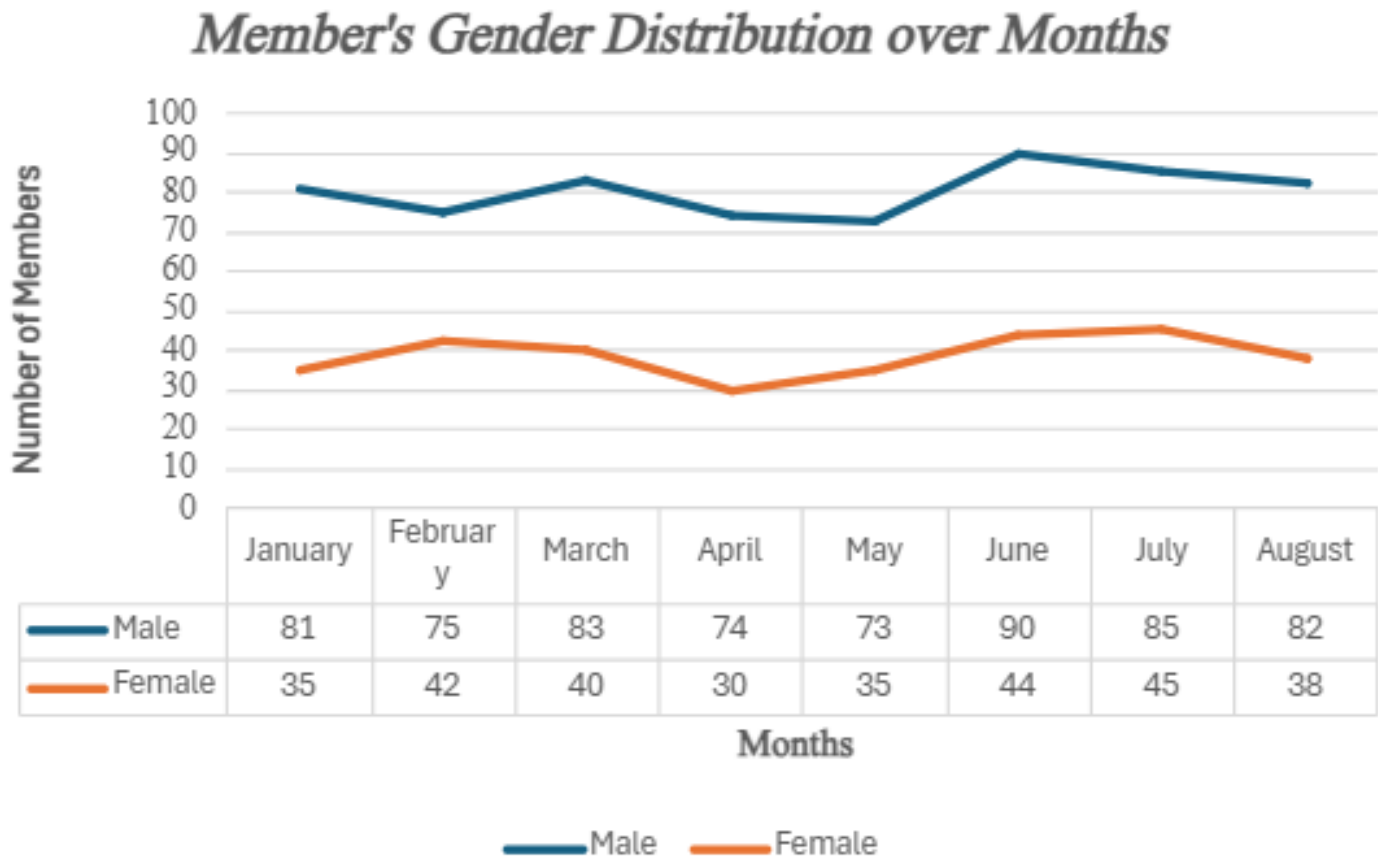


Fig 4. Line chart showing Member's Gender Distribution over months

- **About the Graph:** The above Line chart showing Male vs. Female Membership Distribution over a period of 8 months. This graph indicates how many females and how many males are included in the total members, as well as which gender has fewer individuals over a period of 8 months.
- **Findings:** As per data provided it has been observed that number of male members are always high as compared to female members in all the months starting from January 2024 to August 2024.

Gender Distribution of Members	
Male	Female
63%	37%

Male vs. Female membership ratio

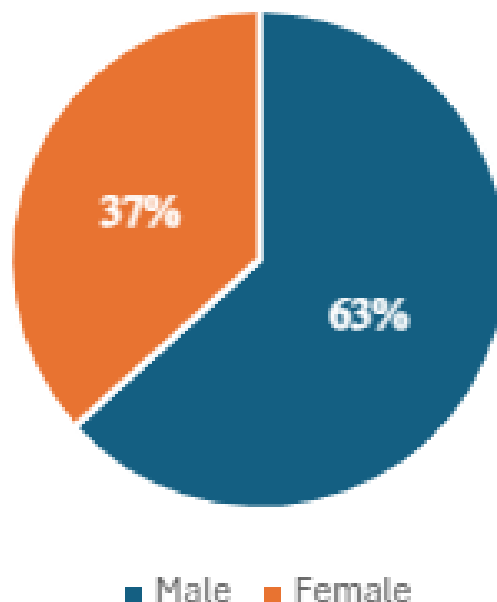


Fig 5. Pie – chart showing Member’s Gender Distribution for the month (August)

- **About the Chart:** The chart distinguishes the ratio between male and female members in the gym during the month of August. It highlights gender imbalance, indicating which gender is more dominant in membership numbers.
- **Findings:** Number of male members are high as compared to female members.

4. Monthly Membership Fluctuations



Fig 6. Line chart representing Members with monthly membership subscription

- **About the Graph:** Above line chart represents members with monthly membership subscription over the period of 8 months. This graph indicates in which months the monthly members are higher in numbers and in which months they are lower in numbers.
- **Findings:** Membership increases in March, June and July because schools and colleges vacations as school exams end in March and students showcase their interests in the gym, and college exams ends in June and July.

5. Equipment Availability

Time Slot	Area/Zone	Equipment in Use	Number of Members	Comments / Feedback	Ratings (on scale 0 to 5) 0: worst and 5: good
4:00 AM - 5:00 AM	Cardio	Treadmills, Cycles, Skipping	2	All Good	5
5:00 AM - 6:00 AM	Cardio, Free Weights	Treadmills, Squat Rack, Dumbbells	1	All Good	5
6:00 AM - 7:00 AM	Cardio, Weight Machines	Treadmills, Rowing Machines, Benches	18	Crowded, members waiting for equipment	3
7:00 AM - 8:00 AM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	16	High demand, long wait time for benches	2
8:00 AM - 9:00 AM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	21	Crowded, members waiting for equipment	3
9:00 AM - 10:00 AM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	4	All Good	5
4:00 PM - 5:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	7	Moderate Crowd but wait times observed	4
5:00 PM - 6:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	2	All Good	5
6:00 PM - 7:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	10	High demand, long wait time	2
7:00 PM - 8:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	21	High demand, long wait time	2
8:00 PM - 9:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	12	Moderate Crowd but wait times observed	4
9:00 PM - 10:00 PM	Cardio, Weight Machines, Free Weights	Treadmills, Squat Rack, Dumbbells, Rowing Machines, Benches	6	Moderate Crowd but wait times observed	4

Fig 7. Table for Equipment used across time slots

- **About the Table:** Above table represents the equipment used across time slots. This table indicates which equipment is used more at certain times and less at other times, resulting in an increase in members' waiting time.
- **Findings:** The waiting period for members increases due to the equipment primarily used during peak hours, leading to an increase in the dropout rate of memberships.

Interpretation of Results

1. Membership Trends and Retention

Membership Subscription Type Distribution

- **Findings:** Many members prefer the monthly membership subscription type, and most dropouts also occur among members with the monthly subscription type.
- **Interpretation:** New members want quick results however, when they don't see immediate results, they get frustrated and quit.

Member Retention Patterns

- **Findings:** Members with annual or quarterly membership subscriptions demonstrate higher retention rates.
- **Interpretation:** People with annual and quarterly subscriptions show higher retention rates than monthly membership subscriptions because they are more dedicated to their goals.

2. Overcrowding Analysis

- **Findings:** Gym experiences overcrowding during certain periods of time in morning (6 AM – 9 AM) and certain period of times in evening (6 PM – 9 PM) and off-peak hours are (4 AM – 6 AM), (9 AM – 10 AM) in the morning and (4 PM – 6 PM), (9PM – 10 PM) in the evening.
- **Interpretation:** Peak hours of the day: 6 AM – 9 AM and 6 PM – 9 PM as many people want to exercise before going to the office or after returning from work. The most crowded time is typically between 7 pm and 8 pm in the evening, as busy women during the day and late-night diners tend to come during this time, along with some couples or siblings who also want to exercise.

3. Gender Distribution

- **Findings:** As per data provided it has been observed that number of male members are always high as compared to female members in all the months starting from January 2024 to August 2024.
- **Interpretation:** Females are less in comparison to males because they are busy with household work and are less dedicated to fitness.

4. Monthly Membership Fluctuations

- **Findings:** Membership increases in March, June and July while February and October see declines.
- **Interpretation:** In March, June, and July, membership increases as school and college vacations occur; school exams conclude in March, leading students to show interest in the gym, while college exams end in June and July. These fluctuations highlight the need for seasonal marketing campaigns and short-term retention strategies during peak months.

5. Equipment Availability

- **Findings:** The waiting period for members increases due to the equipment primarily used during peak hours, leading to an increase in the dropout rate of memberships.
- **Interpretation:** The lack of sufficient equipment during busy times diminishes the gym experience, particularly for new or less active members. Investing in additional equipment for high-demand areas can enhance the member experience.

Recommendations

A. Recommendations for Member Retention and Acquisition

1. **Initiate Loyalty Programs:**
A loyalty program for monthly members, such as discounts on fees, discounts on trainer fees, or a 5-day extension if they have been a continuous member for the last 6 months. The short-term members will remain engaged for a longer period due to this program, and it will be helpful in member retention.
2. **Launch Targeted Seasonal Promotions:**
Campaigns during peak months (March, June and July) to attract school and college students with special discounts and free trials for students, invite local celebrities or fitness icons. which will assist in member acquisition.
3. **Launch Collaboration program:**
Collaborate with local businesses and provide discounts on fees to their employees if they register for a long period and if more people register, which will help in member acquisition.
4. **Flexible Membership Options:**

Introduce options like pay per visit, which means you only need to pay for the day you come, 100 rupees per day. This is for those who want to join temporarily for a few days, and this type of option will help the business in generating additional profit.

5. **Launch Health and Wellness Workshops:**
Organize workshops on nutrition, physical and mental fitness with fitness icons, keep these workshops free for everyone and hold them in parks and similar venues to gather as many people as possible. These workshops will assist in member acquisition.
6. **Free Trial Periods for New Members:**
Attract new members by offering a one-week free trial and also arrange a demo class with a trainer so that they can experience the gym facilities.
7. **Member Feedback and Support:**
Regularly collect feedback from the members regarding the difficulties they are facing and based on this, we will provide the necessary support.

B. Recommendations for managing overcrowding at certain period of times

1. **Off-Peak Hour Discounts on fees:**
To reduce overcrowding during peak hours, offer the discounted rates for those members who come during off-peak hours. This will prevent the gym from becoming overcrowded at certain period of time and will also help in member retention, as those who want to change gyms due to overcrowding will retain.
2. **Offer Group Classes During Off-Peak Hours:**
Schedule group workout classes during off-peak hours, such as yoga and Zumba, to attract those individuals who are seeking such exercise sessions.
3. **Hiring Additional Staff for Peak Hours:**
During peak hours, hire additional trainers and support staff who will assist members and ensure smooth operations. This will enhance customer service and provide members with personalized attention, which will also aid in member retention.
4. **Enhance Zone Utilization:**
Redesign the gym layout using data on overcrowding and improve the efficiency of the area. For instance, relocating high-demand equipment to less utilized areas can reduce congestion and enhance flow.

5. Expand Operating Hours of Gym:
Extend the gym operating hours to allow more flexibility for members who prefer to avoid peak hours and will also help in managing the crowd at certain periods of time.